

A Robust HECC-based Authentication and Key Agreement for UAV-enabled SAR Networks

Ruiyang Huang, *Student Member, IEEE*, Ning Gao, *Member, IEEE*, Qiang Ni, *Senior Member, IEEE*, and Shi Jin, *Fellow, IEEE*

Abstract—Unmanned aerial vehicles (UAVs) have become an indispensable tool in search-and-rescue (SAR) missions, playing a crucial role in locating victims and delivering essential supplies in disaster-stricken areas. However, due to communications in open channels, these unignorable security issues including unauthorized data access, data tampering and leakage, potentially disturb, or even damage the SAR missions. The authentication and key agreement (AKA) technologies are designed to try to allow only verified legal entities can access the data securely with a negotiated session key. Despite the advantages of AKA, existing UAV-specific AKA solutions are unable to resist specific attacks which threaten the security of authentication and session key, and are plagued from the unsatisfactory performance given limited resources of UAVs. To address these issues, based on hyperelliptic curve cryptography (HECC), we propose a robust AKA for UAV-enabled SAR network and preserve the desiring performance as well. With detailed security analysis, we demonstrate the robustness of our scheme under the eCK Adversary Model. Furthermore, the performance analysis shows the efficiency and utility of our design in terms of computational overhead and communication cost, in which it indicates that our scheme can be applied to the UAV-enabled SAR network.

Index Terms—Unmanned aerial vehicle (UAV), Authentication and key agreement (AKA), Hyper-elliptic curve cryptography (HECC), eCK Adversary Model

I. INTRODUCTION

WITH the rapid development of UAVs driven by advancements in technologies such as intelligent systems, communication systems, and sensing systems, UAVs have become an increasingly vital tool [1] in urban search-and-rescue (SAR), offering new opportunities [2] to enhance the rescue efficiency and capabilities of decision-making [3]. As illustrated in Figure 1, UAVs can quickly reach the location affected by the disaster, and further captures real-time imagery data via the installed advanced sensors such as high-definition cameras and thermal imaging devices [4]. The acquired real-time disaster assessment data are transmitted back to the ground core network (GCN). Upon receiving the information, the GCN coordinates nearby rescue resources, which collaborate with the internet of vehicles (IoV) to reach

the disaster site for time-critical rescue operations. Thanks to this multimodal integrated sensing and communications (ISAC) technology, UAV-enabled SAR network significantly reduces search times while increasing accuracy for detecting the trapped individuals [5], [6].

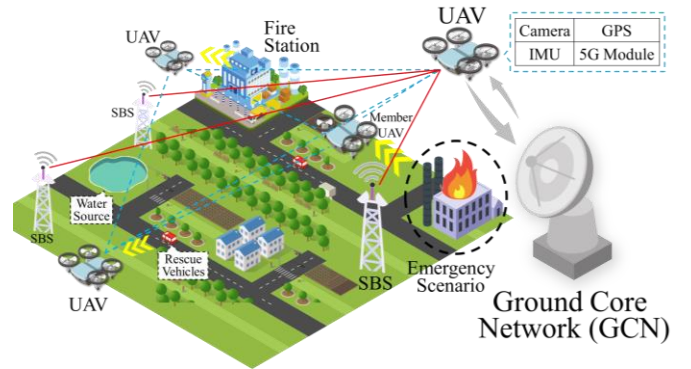


Fig. 1. UAV-enabled SAR network.

Open wireless channels offer critical advantages for UAV urban rescue communication systems, including ubiquitous coverage for seamless connectivity in dynamic environments, cost-effective deployment without physical infrastructure constraints, scalability to support massive device onboarding, and real-time adaptability for time-sensitive applications like UAV swarms, making them indispensable for rapid-response systems. However, UAV communications in urban emergency response rely on open wireless channels, which expose inherent vulnerabilities due to broadcast transmission and unsecured signal propagation [7], including identity spoofing, man-in-the-middle attacks, and other attacks [8]-[12]. For example, in 2017, the Chengdu Shuangliu Airport drone intrusion involved attackers hijacking control signals using reverse-engineered radio protocols, causing flight delays. Additionally, in 2023, a DJI vulnerability allowed man-in-the-middle (MITM) attacks to intercept unencrypted video streams and flight logs, leading to data breaches or mission record tampering.

The technologies of authentication and key agreement (AKA) provide a mechanism to ensure the security of UAV communications by enabling mutual authentication and generating session keys for encryption, mitigating the risks of unauthorized access and tampering. In recent years, there are many AKA schemes for UAV identity authentication and communication security [13]-[16], [18], [23]. However, related SAR-oriented AKA schemes expose critical limitations in security and performance including resistance to limited attacks and high computational costs from resource-intensive operations strain resource-constrained UAVs, causing unsafety

R. Huang and N. Gao are with the School of Cyber Science and Engineering, Southeast University, Nanjing 210096, China (e-mail: ryhuang_572@seu.edu.cn; ninggao@seu.edu.cn).

Q. Ni is with the School of Computing and Communications, Lancaster University, Lancaster LA1 4WA, U.K. (e-mail: q.ni@lancaster.ac.uk).

S. Jin is with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China (e-mail: jinshi@seu.edu.cn).

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and latency in emergencies.

Concentrating on these problems, in this paper, we propose a novel identity authentication scheme based on the hyper-elliptic curve cryptography (HECC) [17], which demonstrates both robust security and satisfactory operational performance. The contributions of this study can be listed below.

- Firstly, we analyze Muhammad et al.'s work [18] and show their scheme remains vulnerable to certain threats, including ephemeral secret leakage (ESL) attacks and Type-I node capture attacks [19], which pose a threat to the security of session keys, especially in dynamic, adversarial environments.
- Secondly, to address the loopholes above, a robust and efficient AKA scheme for UAV-SAR network is designed. Our proposed solution supports mutual authentication between UAVs, and generate a shared session key to protect the security of the transmitted data. Next, on the one hand, the formal security analysis has been proven that the session key can be semantic security, and the heuristic analysis has been shown that the designed scheme can be against the attacks under the eCK adversary model.
- Lastly, to show the efficiency of the proposed scheme, from the view of the communication cost and computation cost, we describe the comparative performance analysis between the proposed scheme and the benchmark schemes. The comparison results indicate that our scheme has the least communication cost, with the second lowest computational cost, while providing higher security than the scheme with the least computational cost.

The rest part of this study is organized as follows. Section II provides a review of related works. Section III presents the system model and in Section IV, a detailed cryptanalysis of Muhammad et al.'s scheme [18] is conducted to highlight its vulnerabilities. Section V and VI introduces our proposed scheme with a comprehensive security analysis. Finally, Section VII evaluates the performance of the proposed scheme, and Section VIII concludes the study.

II. RELATED WORKS

In recent years, with the rapid advancement of unmanned aerial vehicle (UAV) technology, numerous AKAs have been proposed to mitigate the risks associated with unauthorized access, data breaches, and malicious attacks [20]. These schemes primarily address the security challenges in identity authentication and session key establishment.

Specifically, the authors Pu introduced the PMPD2D scheme [21], which facilitates mutual authentication and session key establishment between UAVs. However, this scheme lacks comprehensive consideration for the entire communication lifecycle, leaving potential vulnerabilities during extended operations. Zhang et al. proposed a lightweight AKA scheme [22] that reduces computational overhead, but its security analysis is incomplete, leaving uncertainties about its robustness. Similarly, Gope et al. designed a privacy-preserving AKA scheme [23] utilizing edge computing, but the high computational complexity of the solution poses challenges for

implementation in resource-constrained UAV environments.

Following, researchers have designed AKA schemes for specific application scenarios. Jan et al. developed an AKA scheme [24] for civilian UAVs, however, it cannot adequately address privacy protection, potentially exposing sensitive operational data. Khan et al. proposed a certificate-based access control and AKA scheme [18], but its scalability is limited, making it less suitable for large-scale UAV networks. Zhang et al. presented an anonymous AKA scheme [25] for intelligent UAVs, but its reliability and practical deployment remain unverified.

Some works have explored PUF-based lightweight AKA schemes to enhance efficiency. Wang et al. introduced a multi-factor and PUF-based AKA scheme [26] that significantly improved authentication speed. Ever et al. proposed a lightweight AKA scheme [27] for mobile sink nodes, but its security analysis requires further improvement to establish its robustness against advanced attacks.

Furthermore, blockchain-based cross-domain AKA scheme has garnered attention for their potential to secure complex UAV networks. Feng et al. proposed a blockchain-based cross-domain AKA scheme [28], enabling secure registration, authentication, and auditing of UAVs. However, its real-time performance and efficiency remain areas for improvement. Tanveer et al. introduced a three-factor AKA scheme [29] to enhance security, but its high computational overhead makes it unsuitable for resource-constrained UAV systems.

Overall, these AKA schemes provide effective security measures for UAV communication but exhibit several limitations. Specifically, most schemes focus on a single security objective, such as identity authentication or session key establishment, without addressing the entire communication cycle comprehensively. Additionally, the high computational complexity and communication cost of some schemes hinder their applicability in resource-constrained environments.

III. PRELIMINARIES

A. System Model

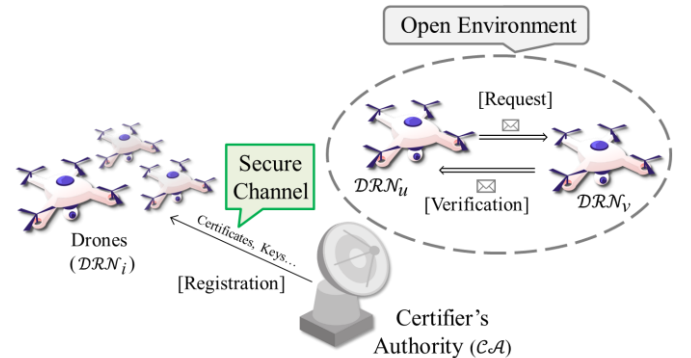


Fig. 2. System model.

The system model in our scheme is mainly composed of two entities: a trusted certifier's authority ($\mathcal{CA}$) and different series of drone nodes ($\mathcal{DRN}$'s). In general, $\mathcal{CA}$ is responsible for the distribution of identity information and the computation of authentication certificates; while the $\mathcal{DRN}$'s, except for the

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registration when they need to contact with $\mathcal{CA}$, are distributed in suitable areas to engage in their respective tasks and interact directly with each other without the need to establish a session through $\mathcal{CA}$. For the sake of clarity, all the symbols and their descriptions used in our system are illustrated in Table I.

TABLE I
DESCRIPTIONS OF SYMBOLS IN OUR SCHEME

Symbols	Descriptions
$\mathcal{CA}$	trusted certifier's authority
$\mathcal{DRN}_i$	the i^{th} drone
ID_i	unique identity of $\mathcal{DRN}_i$
$\mathcal{Cr}_i$	certificate of $\mathcal{DRN}_i$
(a_i, A_i)	private and public key of $\mathcal{DRN}_i$
$(d_{\mathcal{CA}}, w_{\mathcal{CA}})$	private and public key of the certifier's authority
D	the generator of the hyper-elliptic curve
$\mathcal{H}(\cdot)$	the secure hash function
$\mathcal{R}_i$	random salt of $\mathcal{DRN}_i$
(ω_i, Ω_i)	$\mathcal{DRN}_i$'s temporary private and public key of the session
ω_i^*	$\mathcal{DRN}_i$'s temporary salt of the session
A_i	$\mathcal{DRN}_i$'s signature for the session
Ψ_i	the i^{th} message in the public channel
$\mathcal{SK}_{uv}$	shared key of the session between $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$
$\mathcal{E}_{key}(\cdot)/\mathcal{D}_{key}(\cdot)$	encryption/decryption with the key
$X Y$	the concatenate operation of string X and Y
$\Rightarrow/\rightarrow$	public/private channels in the open/secure environment

We illustrate the procedure of the authentication and key agreement in such a scenario. In the initial phase, $\mathcal{CA}$ generates its own system parameters, which mainly include the long-term private key, and the corresponding public key. When a drone (e.g., $\mathcal{DRN}_i$) registers with $\mathcal{CA}$, it sends a registration request to the $\mathcal{CA}$, which then assigns it an identity (ID_i) and sends the generated private key, its corresponding public key, and other values to $\mathcal{DRN}_i$.

In the following phase, one of the drones $\mathcal{DRN}_u$ initiates a session request to another drone $\mathcal{DRN}_v$, selects a temporary session private key and a one-time random number (nonce) for this session, and computes some parameters based on public-key cryptosystems. Then, $\mathcal{DRN}_u$ uses its own private key and certificate to make a signature. Further, $\mathcal{DRN}_u$ sends request message with signature to $\mathcal{DRN}_v$. After receiving the session request, $\mathcal{DRN}_v$ decrypts the message using its own private key and the other's public key to confirm the identity and gain the parameters. Similarly, $\mathcal{DRN}_v$ will send the same request along with the parameters corresponding to the shared session key to $\mathcal{DRN}_u$. Eventually, $\mathcal{DRN}_u$ receives parameters from $\mathcal{DRN}_v$ and calculates the temporary shared session key for both $\mathcal{DRN}$ s, at this time, both UAVs enable the process in AKA.

B. Threat Model

In the design of the authentication protocols, the extended

Canetti-Krawczyk (eCK) model proposed by LaMacchia et al. [30] is used for evaluating the security of cryptographic protocols, which aims to take into account possible attack methods and attacker's capabilities more comprehensively. Specifically, it has been proved in reference [31] that the adversary's greater ability than the Dolev-Yao (DY) model. Therefore, our proposed scheme based on the eCK model will exhibit more robust security compared to the conventional AKA schemes based on the DY model. In this more complete threat model, Attacker $\mathcal{A}$ can be entitled to the six capabilities in Table II [32].

IV. CRYPTANALYSIS OF MUHAMMAD ET AL.'S SCHEME

In this section, we focus on the security of Muhammad et al.'s scheme [18] and then demonstrate the three security threats found from the drone access control process in Muhammad et al.'s scheme. Given the limited space, in the following, we only show three attack processes, and the description of drone access control process can be found in Muhammad et al.'s [18], pp 4842].

A. No Resistance to ESL Attacks

In the AKA scheme, an ESL attack, or "Ephemeral Secret Leakage" attack, refers to a scenario in which an adversary gains access to the temporary private key of one of the two communication parties. Once the adversary has obtained this type of key, the session key $\mathcal{K}_{uv}$ in Muhammad et al.'s scheme can be revealed by the adversary under the following circumstances.

Attack target: get the session key $\mathcal{K}_{uv}$.

Attacker's capability:

- Obtain $\mathcal{DRN}_u$'s temporary private key ε_u or $\mathcal{DRN}_v$'s temporary private key ε_v ;
- Eavesdrop these messages from the public channel
 $\Psi_1 = (NON_{eu}, \Omega_u, A_u, b_u, \mathcal{X}_u, \mathcal{Cr}_u)$,
 $\Psi_2 = (NON_{ev}, \Omega_v, A_v, b_v, \mathcal{X}_v, \mathcal{Cr}_v, \mathcal{VK}_{uv}, ID_v)$,
 $\Psi_3 = (NON_{ev}^*, \mathcal{VK}_{uv}^*)$.

Attack path:

- 1) Obtain $\mathcal{DRN}_v$'s temporary private key ε_v ;
- 2) Base on the relation relevant to A_v , and compute

$$a_v = \frac{A_v - \mathcal{Cr}_v}{\mathcal{H}(\mathcal{X}_v || \Omega_v || \mathcal{Cr}_v || b_v || NON_{ev})} - \varepsilon_v;$$

- 3) Use a_v to decrypt $(N_u, ID_u) = \mathcal{D}_{a_v}(NON_{eu})$ and get ID_u ;
- 4) Use a_v to compute $\mu_{uv} = a_v \cdot b_u = a_v a_u \cdot D$;
- 5) Use ε_v to compute $\gamma_{uv} = \varepsilon_v \cdot \Omega_u = \varepsilon_v \varepsilon_u \cdot D$;
- 6) Compute $\mathcal{K}_{uv} = \mathcal{H}(ID_u || NON_{ev} || \gamma_{uv} || \mu_{uv} || ID_v)$.

Time complexity: $O(2T_h + 2T_p + T_D)$. Here, T_h , T_p , and T_D denote the time for hash function, additive group's

TABLE II
THREAT MODEL

C.	Description of Attackers' Capacities
C_1	$\mathcal{A}$ can fully control the public channel and then intercept, modify, insert, and delete any messages transmitted in the open channel
C_2	$\mathcal{A}$ can masquerade as a legal UAV to communicate in the open channel
C_3	$\mathcal{A}$ can obtain the long-term private key of the certifier's authority $\mathcal{CA}$
C_4	$\mathcal{A}$ can acquire previous session keys between the two $\mathcal{DRN}$ s
C_5	$\mathcal{A}$ can learn one side of the $\mathcal{DRN}$ s' temporary private key in a session when considering the system's eventual failure
C_6	$\mathcal{A}$ can break some UAV nodes and then extract the stored sensitive data, and even control the broken node to join the next communication interaction

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multiplicative operation, and decryption, respectively.

B. No Resistance to Node Capture Attacks

The current research by Wang [33] on node capture attacks describes the Type-I attack, in which the adversary compromises the UAV node's secret value a_u and a_v and eavesdrops on messages from the open channel. This allows the adversary to leak the session key $\mathcal{K}_{uv}$ of the two communication parties. Furthermore, the scheme proposed by Muhammad et al. is susceptible to a Type-I node capture attack.

Attack target: Get the session key $\mathcal{K}_{uv}$.

Attacker's capability:

- Capture and restore $\mathcal{DRN}_u$'s memory ($ID_u, \mathcal{C}r_u, A_u, a_u, \mathcal{X}_u$), or $\mathcal{DRN}_v$'s memory;
- Eavesdrop Ψ_1, Ψ_2, Ψ_3 from the public channel.

Attack path 1:

- 1) Obtain $\mathcal{DRN}_v$'s temporary private key a_v ;
- 2) Base on the relation relevant to A_v , and compute
$$\varepsilon_v = \frac{A_v - \mathcal{C}r_v}{\mathcal{H}(\mathcal{X}_u || \Omega_u || \mathcal{C}r_u || b_v || NON_{eu})} - a_v;$$
- 3) Use a_v to decrypt $(N_u, ID_u) = \mathcal{D}_{a_v}(NON_{eu})$ and get ID_u ;
- 4) Use a_v to compute $\mu_{uv} = a_v \cdot \mathcal{E}_u = a_v a_u \cdot D$;
- 5) Use ε_v to compute $\gamma_{uv} = \varepsilon_v \cdot \Omega_u = \varepsilon_v \varepsilon_u \cdot D$;
- 6) Compute $\mathcal{K}_{uv} = \mathcal{H}(ID_u || NON_{eu} || \gamma_{uv} || \mu_{uv} || ID_v)$.

Attack path 2:

- 1) Obtain $\mathcal{DRN}_u$'s temporary private key a_u ;
- 2) Base on the relation relevant to A_u , and compute
$$\varepsilon_u = \frac{A_u - \mathcal{C}r_u}{\mathcal{H}(\mathcal{X}_u || \Omega_u || \mathcal{C}r_u || b_u || NON_{eu})} - a_u;$$
- 3) Use a_u to compute $\mu_{uv}^* = a_u \cdot \mathcal{E}_v = a_v a_u \cdot D$;
- 4) Use ε_u to compute $\gamma_{uv}^* = \varepsilon_u \cdot \Omega_v = \varepsilon_v \varepsilon_u \cdot D$;
- 5) Compute $\mathcal{K}_{uv}^* = \mathcal{H}(ID_u || NON_{eu} || \gamma_{uv}^* || \mu_{uv}^* || ID_v)$;
- 6) Compute $\mathcal{V}\mathcal{K}_{uv}^* = \mathcal{H}(NON_{eu} || \mathcal{K}_{uv}^*)$, and check if $\mathcal{V}\mathcal{K}_{uv} \stackrel{?}{=} \mathcal{V}\mathcal{K}_{uv}^*$.

Time complexity: $O(2T_h + 2T_p + T_D)$ and $O(3T_h + 2T_p)$.

V. THE PROPOSED SCHEME

This section is committed to presenting our proposed scheme, which leverages the HECC and employs a one-way hash function. Our proposed scheme comprises four main steps: Setup, Drone Registration, Drone Access Control Process, and the New Drone Addition Phase. A detailed explanation of each step is provided below:

A. Set-up Phase of Certifier's Authority

The set-up phase is executed by the trustworthy certifier's authority $\mathcal{CA}$. Its main task is to generate private and public keys. Furthermore, a set of public parameters is prepared in this step. To perform such a calculation, the following sequence needs to be followed:

- 1) Firstly, the trusted certification authority $\mathcal{CA}$ selects $d_{\mathcal{CA}}$ as its private key from the dataset $\{1, 2, 3, \dots, n-1\}$.
- 2) It generates the public key as follows:
$$w_{\mathcal{CA}} = d_{\mathcal{CA}} \cdot D,$$
where " D " is the generator of the hyperelliptic curve.
- 3) Choose the hash function $\mathcal{H}$, which has the ability to avoid

collisions and is characterized by irreversibility.

- 4) Choose the common parameter set as $\eta = (d_{\mathcal{CA}}, D, n, \mathcal{H})$ and publish it.

B. UAV Registration Phase

The drone registration step considers a deployed drone $\mathcal{DRN}_u$, assuming that the authorized body $\mathcal{CA}$ wants to register it offline. The process proceeds in the following order:

- 1) For $\mathcal{DRN}_u$, an identity ID_u and a private key a_u are obtained from the dataset $\{1, 2, 3, \dots, n-1\}$.
- 2) $\mathcal{CA}$ uses the following relationship to calculate the public A_u :

$$A_u = a_u \cdot D.$$

- 3) $\mathcal{CA}$ selects $\mathcal{E}_u$ from the dataset $\{1, 2, 3, \dots, n-1\}$ and compute the value of $\mathcal{X}_u$ as:

$$\mathcal{X}_u = (\mathcal{E}_u + a_u) \cdot D.$$

- 4) $\mathcal{CA}$ calculates the certificate for ID_u :
$$\mathcal{C}r_u = d_{\mathcal{CA}} + (\mathcal{E}_u + a_u) \mathcal{H}(ID_u || \mathcal{X}_u).$$
- 5) In the last step, $\mathcal{CA}$ puts the set $(ID_u, \mathcal{C}r_u, A_u, a_u, \mathcal{X}_u)$ into $\mathcal{DRN}_u$ in memory.

C. UAV Access Control Process

Suppose two UAVs $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$ wish to communicate with each other through the AKA mechanism. To carry out this process, there are three stages:

Phase # 1: If $\mathcal{DRN}_u$ wants to establish a shared session key with $\mathcal{DRN}_v$, the operation is performed as follows:

- 1) First, $\mathcal{DRN}_u$ selects different ω_u and ω_u^* from $\{1, 2, 3, \dots, n-1\}$ and performs the following calculation:

$$\Omega_u = \omega_u \cdot D,$$

$$\Omega_u^\circ = (\omega_u + \omega_u^*) \cdot D.$$

- 2) Select a fresh nonce N_u for this session.
- 3) Yield the signature A_u for this session on ω_u and ω_u^* according to the following equation:

$$A_u = \mathcal{C}r_u + \mathcal{H}(ID_u || \mathcal{C}r_u || A_u || \mathcal{X}_u || N_u)(\omega_u + \omega_u^* + a_u).$$

- 4) Finally, $\mathcal{DRN}_u$ sends the message $\Psi_1 = \{E_{A_v}(ID_u, N_u, A_u, \mathcal{C}r_u, \mathcal{X}_u, \Omega_u, \Omega_u^\circ, A_u)\}$ to $\mathcal{DRN}_v$ over the open network, here the public key A_u of $\mathcal{DRN}_v$ is considered to be known to $\mathcal{DRN}_u$.

Phase # 2: Then, after receiving the message Ψ_1 from $\mathcal{DRN}_u$, $\mathcal{DRN}_v$ performs the following calculation:

- 1) First, decrypt the message Ψ_1 using the private key a_v to get $ID_u, N_u, A_u, \mathcal{C}r_u, \mathcal{X}_u, \Omega_u, \Omega_u^\circ, A_u$. Meanwhile, use the formula to calculate the temporary public key Ω_u^* :

$$\Omega_u^* = \Omega_u^\circ - \Omega_u.$$

- 2) Check the certificate of the source drone by applying the following conditions:

$$w_{\mathcal{CA}} + \mathcal{H}(ID_u || \mathcal{X}_u) \cdot \mathcal{X}_u \stackrel{?}{=} \mathcal{C}r_u \cdot D.$$

- 3) Verify the following conditions:
$$\mathcal{C}r_u \cdot D + \mathcal{H}(ID_u || \mathcal{C}r_u || A_u || \mathcal{X}_u || N_u) \cdot (\Omega_u^\circ + A_u) \stackrel{?}{=} A_u \cdot D.$$
- 4) If the signature is valid, further calculations are performed:

$$\Omega_v^\circ = (\omega_v + \omega_v^*) \cdot D.$$

Here ω_v and ω_v^* belong to the dataset $\{1, 2, 3, \dots, n-1\}$.

- 5) Calculate the signature A_v using the following relation:

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$$A_v = \mathcal{C}r_v + \mathcal{h}(ID_v || \mathcal{C}r_v || A_v || \mathcal{X}_v || N_v)(\omega_v + \omega_v^* + \alpha_v).$$

- 6) Perform the following calculations to generate λ_{uv} and μ_{uv} :

$$\lambda_{uv} = \omega_v \cdot \Omega_u = \omega_v \omega_u \cdot D,$$

$$\mu_{uv} = \alpha_v \cdot A_u = \alpha_v \alpha_u \cdot D.$$

- 7) At this point, $\mathcal{SK}_{uv}$ can be calculated based on the relation:

$$\mathcal{SK}_{uv} = \mathcal{h}(\mathcal{C}r_u || N_u || \lambda_{uv} || \mu_{uv} || N_v || \mathcal{C}r_v).$$

- 8) Calculate the value of NON_{ev} and $\mathcal{VK}_{uv}$ based on the relationship:

$$NON_{ev} = \mathcal{E}_{\Omega_u}(N_v, ID_v),$$

$$\mathcal{VK}_{uv} = \mathcal{h}(N_u || N_v || \mathcal{SK}_{uv}).$$

- 9) Finally, $\mathcal{DRN}_v$ passes the message Ψ_2 to $\mathcal{DRN}_u$ over the open network where $\Psi_2 = \{\mathcal{E}_{A_u}(NON_{ev}, A_v, \mathcal{C}r_v, \mathcal{VK}_{uv}), ID_v, \Omega_v^\circ, A_v, \mathcal{X}_v\}$.

Phase # 3: In this stage, $\mathcal{DRN}_u$ receives the message Ψ_2 and performs the following calculation:

- 1) First decrypt the message Ψ_2 using the private key α_u to get $NON_{ev}, A_v, \mathcal{C}r_v, \mathcal{VK}_{uv}, ID_v, \Omega_v^\circ, A_v, \mathcal{X}_v$. Then proceed to decrypt $(N_v, ID_v) = \mathcal{D}_{\omega_u^*}(NON_{ev})$ to get and check the freshness of N_v .
- 2) Check the certificate of the drone $\mathcal{DRN}_v$ by computing the following conditions:
 $w_{\mathcal{CA}} + \mathcal{h}(ID_v || \mathcal{X}_v) \cdot \mathcal{X}_v \stackrel{?}{=} \mathcal{C}r_v \cdot D.$
- 3) Verify the signature by the following conditions:
 $\mathcal{C}r_v \cdot D + \mathcal{h}(ID_v || \mathcal{C}r_v || A_v || \mathcal{X}_v || N_v) \cdot (\Omega_v^\circ + A_v) \stackrel{?}{=} A_v \cdot D.$
- 4) If the signature is valid, the following calculations are performed to obtain λ_{uv} and μ_{uv} :

$$\lambda_{uv} = \omega_u \cdot \Omega_v = \omega_u \omega_v \cdot D.$$

$$\mu_{uv} = \alpha_u \cdot A_v = \alpha_u \alpha_v \cdot D.$$

- 5) At this point, $\mathcal{SK}_{uv}^*$ can be calculated based on the relation:

$$\mathcal{SK}_{uv}^* = \mathcal{h}(\mathcal{C}r_u || N_u || \lambda_{uv} || \mu_{uv} || N_v || \mathcal{C}r_v).$$

- 6) Compute $\mathcal{VK}_{uv}^* = \mathcal{h}(N_u || N_v || \mathcal{SK}_{uv}^*)$, while comparing $\mathcal{VK}_{uv}^* \stackrel{?}{=} \mathcal{VK}_{uv}$ to verify the shared session key $\mathcal{SK}_{uv}^*$.

D. New Drone Addition Phase

This phase involves addition of a new drone $\mathcal{DRN}_{new}$ after establishment of a UAV network. $\mathcal{CA}$ registers the new drone $\mathcal{DRN}_{new}$ in an offline environment. And the process proceeds stepwise as follows:

- 1) For $\mathcal{DRN}_{new}$, an identity ID_{new} and a private key α_{new} are obtained from the dataset $\{1, 2, 3, \dots, n-1\}$.
- 2) $\mathcal{CA}$ uses the following relationship to calculate the public A_{new} :

$$A_{new} = \alpha_{new} \cdot D.$$

- 3) $\mathcal{CA}$ selects $\mathcal{f}_{new}$ from the dataset $\{1, 2, 3, \dots, n-1\}$ and compute the value of $\mathcal{X}_{new}$ as:

$$\mathcal{X}_{new} = (\mathcal{f}_{new} + \alpha_{new}) \cdot D.$$

- 4) $\mathcal{CA}$ calculates the certificate for ID_{new} :

$$\mathcal{C}r_{new} = d_{\mathcal{CA}} + (\mathcal{f}_{new} + \alpha_{new}) \cdot \mathcal{h}(ID_{new} || \mathcal{X}_{new}).$$

- 5) In the last step, $\mathcal{CA}$ puts $(ID_{new}, \mathcal{C}r_{new}, A_{new}, \alpha_{new}, \mathcal{X}_{new})$ into $\mathcal{DRN}_{new}$'s memory.

VI. SECURITY ANALYSIS OF PROPOSED SCHEME

In this section, we present a comprehensive analysis of our proposed scheme, encompassing both theoretical and empirical

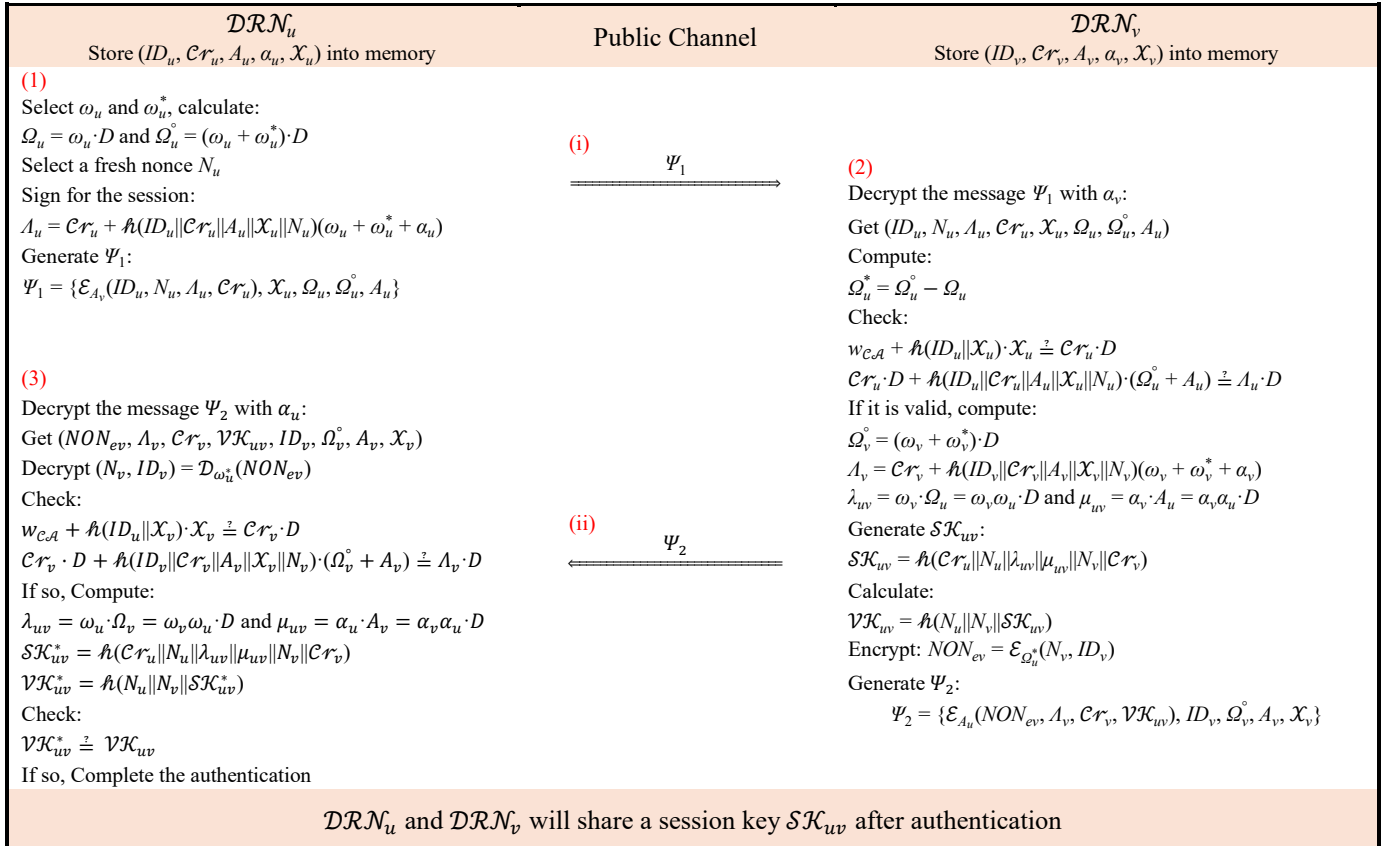


Fig. 3. UAV mutual authentication and key agreement process.

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evaluations. Initially, we demonstrate the semantic security of the session key through a provable proof under the real-or-random (ROR) oracle model, thereby establishing the theoretical foundation of our scheme's security. Subsequently, we conduct heuristic analyses to substantiate the robustness of our scheme against a spectrum of known attacks, showcasing its resilience in practical scenarios.

For the security proof, we give some introductions of basic definitions and then show the detailed formal security analysis.

A. The Basic Definitions

Definition 1: Hyper Elliptic Curve Discrete Logarithm Problem (HECDLP)

The HECDLP is a hard problem in which the attacker attempts to extract f from the relation $L = f \cdot D$, where f is the uniformly selected number from $\{1, 2, 3, \dots, n-1\}$ [36].

Definition 2: Hyper Elliptic Curve Computational Diffie-Hellman Problem (HECCDHP)

The HECCDHP is also a hard problem in which the attacker calculates the value of $xy \cdot D$ when $x \cdot D$ and $y \cdot D$ are known, where x, y are uniformly selected from $\{1, 2, 3, \dots, n-1\}$ [37].

B. Formal Security Analysis

1) The detailed basis: In this section and the subsequent formal proof, the term $\mathbb{C}$ is employed to indicate the proposed scheme. In $\mathbb{C}$, there are three entities: two drones ($\mathcal{DRN}_u$ and $\mathcal{DRN}_v$) and a certifier's authority ($\mathcal{CA}$). Then, the simulator $\mathcal{S}$ initializes a hyper-elliptic curve $HE(F_p)$ over a large prime finite field F_p and selects a q -order additive subgroup $\mathbb{G}$ with a generator D , where the length of the large prime q is a security parameter n . Next, the drones $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$ are provided with their own information.

Subsequently, the three entities will instantiate $\mathcal{DRN}_u$ with instance $\Pi_{\mathcal{DRN}_u}^u, \mathcal{DRN}_v$ with $\Pi_{\mathcal{DRN}_v}^v$, and $\mathcal{CA}$ with $\Pi_{\mathcal{CA}}^{ca}$ respectively. In the absence of a requirement to distinguish between $\Pi_{\mathcal{DRN}_u}^u, \Pi_{\mathcal{DRN}_v}^v$ and $\Pi_{\mathcal{CA}}^{ca}$, any instance may be designated as Π^i . Thereafter, each instance will be simulated as an oracle. In particular, if the input message is valid, incorrect, or null, the oracle's state will manifest as acceptance, rejection, or " $\perp$ (NULL)", respectively.

Based on the threat model, which describes the eCK adversary's capabilities, we define some indispensable items as a basis for a formal proof that will be used to prove the semantic security.

Accepted state: Upon receipt of the last expected protocol message, the instance Π^i receives an accepted state. Meanwhile, throughout the communication, the ordered concatenation of all sent and received messages forms the session identifier of Π^i .

Partnering: Two instances Π^{t1}, Π^{t2} are partnered if Π^{t1}, Π^{t2} both meet: a) in the accepted state, b) by mutual authentication with the shared identical session identifier, and c) Π^{t1}, Π^{t2} are mutual partners in a session.

Adversary: In $\mathbb{C}$, the eCK adversary $\mathcal{A}$ can initiate the query oracles with instance Π^i . The queries that adversary $\mathcal{A}$ can initiate are listed below:

- **Execute** ($\Pi_{\mathcal{DRN}_u}^u, \Pi_{\mathcal{DRN}_v}^v$ and $\Pi_{\mathcal{CA}}^{ca}$). In this query, the

adversary $\mathcal{A}$ participates in the authentication process simulated by $\Pi_{\mathcal{DRN}_u}^u, \Pi_{\mathcal{DRN}_v}^v$ and $\Pi_{\mathcal{CA}}^{ca}$, and can yield the complete communication messages among $\Pi_{\mathcal{DRN}_u}^u, \Pi_{\mathcal{DRN}_v}^v$ and $\Pi_{\mathcal{CA}}^{ca}$.

- **Send** (Π^i, m). $\mathcal{A}$ in this query can initiate an active attack. Specifically, $\mathcal{A}$ submits a valid message m to a participating instance Π^i . Accordingly, the Π^i will give a response to $\mathcal{A}$.
- **SessionKeyReveal** (Π^i). In this query, apart from the session key to be tested, $\mathcal{A}$ can get the other session keys by SessionKeyReveal (Π^i) (and its partner).
- **EphemeralKeyReveal** (Π^i). This query means that the adversary $\mathcal{A}$ can get entities' ephemeral secrets.
- **Corrupt** ($\Pi_{\mathcal{DRN}_u}^u$ and $\Pi_{\mathcal{DRN}_v}^v$). This query means that $\mathcal{A}$ will get the long-term key pair (α_u, A_u) and (α_v, A_v) .
- **Corrupt** ($\Pi_{\mathcal{CA}}^{ca}$). For this query, the secret value $d_{\mathcal{CA}}$ of $\Pi_{\mathcal{CA}}^{ca}$ can be known by $\mathcal{A}$.

Freshness: The confidentiality of a communication is ensured if the adversary $\mathcal{A}$ is unable to discern the session key between the communicating parties $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$ by means of the SessionKeyReveal query.

Test: This is employed to illustrate the semantic security of the session key, $\mathcal{SK}_{uv}$, and the fact that the adversary, $\mathcal{A}$, is only capable of querying the session key once. In light of the aforementioned description of $\mathbb{C}$, it can be inferred that the related instance Π^i can only be $\Pi_{\mathcal{DRN}_u}^u$ and $\Pi_{\mathcal{DRN}_v}^v$. In accordance with the formal definition, if the specified instance (Π^i) has not yet established a session key, or if Π^i is not currently considered fresh, or if the Test (Π^i) query has been conducted previously, the query will immediately conclude and directly output a null value ($\perp$). Conversely, if the aforementioned conditions are not met, the oracle will randomly select a value b , which will be either 0 or 1. If $b = 0$, the Test (Π^i) query will return the actual session key ($\mathcal{SK}_{uv}$). Otherwise, the query will output a random string of the same length as the actual $\mathcal{SK}_{uv}$.

2) Semantic Security: Let $\text{Succ}(\mathcal{A})$ denote the advantage of $\mathcal{A}$ correctly guessing b^* of b . The advantage of $\mathcal{A}$ breaking the semantic security of $\mathcal{SK}_{uv}$ in $\mathbb{C}$ can be defined as follows:

$$\text{Adv}_{\mathbb{C}}^{\mathcal{A}} = 2\Pr[\text{Succ}(\mathcal{A})] - 1. \quad (1)$$

Theorem: Let $\text{Adv}_p^{\text{HECDL}}(n)$ and $\text{Adv}_p^{\text{HECCDH}}(n)$ be defined as the advantage of a PPT adversary $\mathcal{A}$ solving the HECDL and HECCDH hardness problems, respectively. Based on a sequence of queries with oracle, the advantage $\text{Adv}_{\mathbb{C}}^{\mathcal{A}}$ of a PPT adversary $\mathcal{A}$ breaking the semantic security of $\mathcal{SK}_{uv}$ in $\mathbb{C}$ is less than:

$$\Delta + \frac{q_h^2 + q_s + (q_s + q_e)^2}{2^{t1-1}p}, \quad (2)$$

where $\Delta = 2(\text{Adv}_p^{\text{HECDL}}(n) + \text{Adv}_p^{\text{HECCDH}}(n))$ and q_e (resp., q_s , q_h) denotes number of times $\mathcal{A}$ running Execute-query (resp., Send, Hash-query).

Proof: Here a chain from Game₁ to Game₇ constitutes a proof process. Let Succ_i mean that $\mathcal{A}$ successfully guesses b in Test-query of Game _{i} , ($i = 1, 2, \dots, 7$).

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Game₁: This game simulates a real attack under the random oracle model. In this game, a bit b is chosen at the beginning. Then we have:

$$Adv_{\mathcal{C}}^{\mathcal{A}} = 2 \Pr [Succ_1] - 1. \quad (3)$$

Game₂: The game model maintains a hash list, designated as $\Gamma_{\mathcal{H}}$ [34]. Let us consider the following scenario: $\mathcal{A}$ initiates a hash query, designated as $\mathcal{H}(\gamma)$, and the hash oracle $\Theta_{\mathcal{H}}$, then takes γ in order to retrieve $\Gamma_{\mathcal{H}}$. In the event that a hash value, designated as $\mathcal{H}(\gamma)$, can be retrieved from the hash list, $\Gamma_{\mathcal{H}}$, the hash oracle $\Theta_{\mathcal{H}}$, will respond with the retrieved hash value. In the event that the aforementioned conditions are not met, the hash oracle $\Theta_{\mathcal{H}}$ transmits a randomly generated string ψ to $\mathcal{A}$. Concurrently, the response (γ, ψ) is stored in $\Gamma_{\mathcal{H}}$.

By utilizing the known hash list $\Gamma_{\mathcal{H}}$ in Game₂, the adversary $\mathcal{A}$ attempts to discern the value of b by distinguishing it from a random string. In point of fact, the session key $\mathcal{SK}_{uv} = \mathcal{H}(\mathcal{C}\mathcal{R}_u || N_u || \lambda_{uv} || \mu_{uv} || N_v || \mathcal{C}\mathcal{R}_v)$ is derived from the secret values as follows: $\lambda_{uv} = \omega_u \omega_v \cdot D$, $\mu_{uv} = \alpha_u \alpha_v \cdot D$. Therefore, in the absence of knowledge regarding the secret values α_u and α_v , it is not possible for $\mathcal{A}$ to compute $\mathcal{SK}_{uv}$ and there is no effective method by which they can distinguish the real session key from a random string, other than through guesswork.

Consequently, the probability of adversary $\mathcal{A}$ winning Game₂ does not confer any advantage to $\mathcal{A}$ compared with Game₁, that is to say:

$$\Pr [Succ_1] = \Pr [Succ_2]. \quad (4)$$

Game₃: The following section will present a simulation of an active attack in this game. In order to persuade a participant to accept a forged message, the adversary $\mathcal{A}$ initiates a series of *Send*-query and *Hash*-query operations. In comparison to the preceding two games, $\mathcal{A}$ may gain an advantage by identifying a collision. In other words, should the following collisions occur, Game₃ will be terminated.

(i) It is possible to identify collisions in the hash values, and therefore the probability is $\frac{q_{\mathcal{H}}^2}{2^{l_1+1}}$, where l_1 denotes the length of the output produced by a hash function.

(ii) An additional potential collision may be identified by selecting a pair of random numbers with a probability of $\frac{(q_e + q_s)^2}{2^p}$.

Consequently, we have:

$$|\Pr [Succ_3] - \Pr [Succ_2]| \leq \frac{q_{\mathcal{H}}^2}{2^{l_1+1}} + \frac{(q_e + q_s)^2}{2^p}. \quad (5)$$

Game₄: In this game, $\mathcal{A}$ attempts to ascertain $\mathcal{C}\mathcal{R}_i$, A_i , $\mathcal{SK}_{uv}$, and $\mathcal{V}\mathcal{K}_{uv}$ without utilizing a *Hash*-query. Thus, the following can be inferred:

$$|\Pr [Succ_4] - \Pr [Succ_3]| \leq \frac{q_s}{2^{l_1}}. \quad (6)$$

Game₅: In this game, the adversary $\mathcal{A}$ interacts with the *EphemeralKeyReveal* (Π') oracle and the *SessionKeyReveal* (Π') oracle [35], obtaining outdated session keys $(\mathcal{SK}_{uv})_{outdated}$ and the temporary secrets ω_u and ω_v . Thus, $\mathcal{A}$ attempts to corrupt the authentication. However, $\mathcal{A}$ is unable to reveal ω_i^* and N_i from $\mathcal{DRN}_i$'s memory. Alternatively, $\mathcal{A}$ may attempt to identify a collision in the hash values.

This results in the following inequality:

$$|\Pr [Succ_5] - \Pr [Succ_4]| \leq \frac{q_{\mathcal{H}}^2}{2^{l_1+1}}. \quad (7)$$

Game₆: In this game, the adversary $\mathcal{A}$ interacts with the *Corrupt* ($\Pi_{\mathcal{CA}}^{ca}$) oracle, thereby acquiring the private key $d_{\mathcal{CA}}$ of Certifier's Authority $\mathcal{CA}$. Although $\mathcal{A}$ has gained access to $\mathcal{CA}$, the fact that no PPT solution can be used to break the *HECDL* problem means that $\mathcal{A}$ cannot extract α_i from $\mathcal{C}\mathcal{R}_i$. Therefore, we can conclude that:

$$|\Pr [Succ_6] - \Pr [Succ_5]| \leq Adv_p^{HECDH}(n). \quad (8)$$

Game₇: In the final stage of the game, $\mathcal{A}$ is unable to initiate any queries and must instead attempt to compute the session key $\mathcal{SK}_{uv}$. Similarly, given that no PPT solution can be employed to solve the *HECCDH* hardness problem, $\mathcal{A}$ has a negligible advantage $Adv_p^{HECCDH}(n)$ to obtain $\lambda_{uv} = \omega_u \omega_v \cdot D$ and $\mu_{uv} = \alpha_u \alpha_v \cdot D$. Therefore:

$$|\Pr [Succ_7] - \Pr [Succ_6]| \leq Adv_p^{HECCDH}(n). \quad (9)$$

Now, it is evident that $\mathcal{A}$ does not possess any discernible advantage over $\frac{1}{2}$, and so $\Pr [Succ_7] = \frac{1}{2}$. Consequently, from Eq. (1) to Eq. (9), by employing the triangular inequality, we obtain the following deduction:

$$\begin{aligned} Adv_{\mathcal{C}}^{\mathcal{A}} &= 2 \Pr [Succ_1] - 1 \\ &= 2 \Pr [Succ_7] - 1 - 2(\Pr [Succ_7] - \Pr [Succ_1]) \\ &= 2 \Pr [Succ_7] - 1 - 2 \sum_{i=1}^6 (\Pr [Succ_{i+1}] - \Pr [Succ_i]) \\ &\leq \Delta + \frac{q_{\mathcal{H}}^2 + q_s}{2^{l_1+1}} + \frac{(q_e + q_s)^2}{p}. \end{aligned}$$

Therefore, it can be concluded that the adversary $\mathcal{A}$ has no obvious advantage $Adv_{\mathcal{C}}^{\mathcal{A}}$ in breaking the semantic security of $\mathcal{SK}_{uv}$. $\square$

C. Informal Security Analysis

1) Mutual Authentication: The proposed scheme can provide mutual authentication. Suppose if $\mathcal{DRN}_u$ dispatches the message $\Psi_1 = \{\mathcal{E}_{A_v}(ID_u, N_u, A_u, \mathcal{C}\mathcal{R}_u), \mathcal{X}_u, \Omega_u, \Omega_u^*, A_u\}$ to $\mathcal{DRN}_v$.

After receiving Ψ_1 , $\mathcal{DRN}_v$ will performs the following computations:

- First, it decrypts the message Ψ_1 using the private key α_v to get $(ID_u, N_u, A_u, \mathcal{C}\mathcal{R}_u, \mathcal{X}_u, \Omega_u, \Omega_u^*, A_u)$. Meanwhile, it uses the formula to calculate the temporary public key Ω_u^* :

$$\Omega_u^* = \Omega_u^\circ - \Omega_u.$$

- Check the certificate of the source drone by applying the following conditions:

$$w_{\mathcal{CA}} + \mathcal{H}(ID_u || \mathcal{X}_u) \cdot \mathcal{X}_u \stackrel{?}{=} \mathcal{C}\mathcal{R}_u \cdot D.$$

- Attempt to verify the signature, and to check the relationship, verify the following conditions:

$$\mathcal{C}\mathcal{R}_u \cdot D + \mathcal{H}(ID_u || \mathcal{C}\mathcal{R}_u || A_u || \mathcal{X}_u || N_u) \cdot (\Omega_u^\circ + A_u) \stackrel{?}{=} A_u \cdot D.$$

Consequently, the veracity of the identity of $\mathcal{DRN}_u$ is confirmed by $\mathcal{DRN}_v$.

In a similar manner, upon the transmission of the message, $\mathcal{DRN}_u$ is able to ascertain the veracity of the identity of $\mathcal{DRN}_v$ by utilizing the following method:

- First, $\mathcal{DRN}_u$ decrypts the message Ψ_2 using the private key α_u to get $(NON_{ev}, A_v, \mathcal{C}\mathcal{R}_v, \mathcal{V}\mathcal{K}_{uv}, ID_v, \Omega_v^\circ, A_v, \mathcal{X}_v)$. Then, it proceeds to decrypt $(N_v, ID_v) = \mathcal{D}_{\omega_u^*}(NON_{ev})$ to get and check the freshness of N_v .

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- Check the certificate of the drone $\mathcal{DRN}_v$ by computing the following conditions:

$$w_{\mathcal{CA}} + \mathcal{H}(ID_v || \mathcal{X}_v) \cdot \mathcal{X}_v \stackrel{?}{=} \mathcal{CR}_v \cdot D.$$
- Eventually, $\mathcal{DRN}_u$ verifies the signature, and to check the relationship, verifying the following conditions:

$$\mathcal{CR}_v \cdot D + \mathcal{H}(ID_v || \mathcal{CR}_v || A_v || \mathcal{X}_v || N_v) \cdot (\Omega_v^\circ + A_v) \stackrel{?}{=} A_v \cdot D.$$

In summary, our scheme allows for mutual authentication.

2) Forward Secrecy: If the adversary $\mathcal{A}$ successfully attacks the certifier's authority $\mathcal{CA}$, and obtains $\mathcal{CA}$'s private key $d_{\mathcal{CA}}$, our scheme is still forward secure.

The adversary wants to compute:

$$\mathcal{SK}_{uv} = \mathcal{H}(\mathcal{CR}_u || N_u || \lambda_{uv} || \mu_{uv} || N_v || \mathcal{CR}_v).$$

This requires $\mathcal{A}$ to obtain the values of $\mathcal{CR}_u$, N_u , λ_{uv} , μ_{uv} , N_v , $\mathcal{CR}_v$. However, even if the adversary has been in control of $\mathcal{CA}$'s private key $d_{\mathcal{CA}}$, $\mathcal{A}$ is still unable to restore the value of $\mathcal{CR}_u$ or $\mathcal{CR}_v$ because the value of the random salt $\mathcal{f}_u$ or $\mathcal{f}_v$ can only be extracted from the equation:

$$\begin{aligned} \mathcal{X}_u &= (\mathcal{f}_u + \alpha_u) \cdot D, \\ \mathcal{X}_v &= (\mathcal{f}_v + \alpha_v) \cdot D. \end{aligned}$$

This is equivalent to solving the *HECDLP* at least twice. Similarly, without the values of the private keys α_u and α_v , the other parameters are also unobtainable. Consequently, the proposed scheme is forward secure, in that all preceding session keys remain secure even in the event of a long-term private key compromise at the certifier's authority $\mathcal{CA}$.

3) Man-in-the-Middle Attack: Suppose the adversary $\mathcal{A}$ intends to alter the message Ψ_1 transmitted between the two drones $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$, the message $\Psi_1 = \{\mathcal{E}_{A_v}(ID_u, N_u, A_u, \mathcal{CR}_u), \mathcal{X}_u, \Omega_u, \Omega_u^\circ, A_u\}$ comprises of multiple parameters, some of which are encrypted. Thus, to be successful in its malicious attack, the adversary $\mathcal{A}$ will need to access the values of A_u :

$$A_u = \mathcal{CR}_u + \mathcal{H}(ID_u || \mathcal{CR}_u || A_u || \mathcal{X}_u || N_u)(\omega_u + \omega_u^* + \alpha_u).$$

Here, N_u only can be obtained by decrypting message Ψ_1 with α_v . Similarly, $\mathcal{CR}_u$ and other parameters are encrypted. In other words, $\mathcal{A}$ must have the values of α_u and α_v , which could only be obtained from $A_u = \alpha_u \cdot D$ and $A_v = \alpha_v \cdot D$. Besides, $\mathcal{A}$ needs to calculate $(\omega_u + \omega_u^*)$ by $\Omega_u^\circ = (\omega_u + \omega_u^*) \cdot D$.

Performing such mathematical maneuvers is impracticable since it is equivalent to solving the *HECDLP* three times. As it is difficult, in consequence, $\mathcal{A}$ is impossible to solve the problem, and so the proposed scheme can resist the Man-in-the-Middle Attacks.

4) Drone Impersonation Attack: Assuming adversary $\mathcal{A}$ attempts to impersonate $\mathcal{DRN}_v$ to generate the message:

$$\Psi_2 = \{\mathcal{E}_{A_u}(NON_{ev}, A_v, \mathcal{CR}_v, \mathcal{VK}_{uv}), ID_v, \Omega_v^\circ, A_v, \mathcal{X}_v\}.$$

Such attempts, in turn, would require a great deal of computation to produce the variable A_v and $\mathcal{VK}_{uv}$. The mathematical formulae involved are as follows:

$$i) \mathcal{VK}_{uv} = \mathcal{H}(N_u || N_v || \mathcal{SK}_{uv})$$

Here, to approach the value of $\mathcal{VK}_{uv}$, $\mathcal{A}$ is required to compute the value of N_u and $\mathcal{SK}_{uv}$. For the reason that N_u is encrypted by A_v in the message Ψ_1 in the open channel, only when $\mathcal{A}$ have the private key of $\mathcal{DRN}_v$, can he get the proper value of N_u to compute $\mathcal{VK}_{uv}$, which means $\mathcal{A}$ is need to solve the *HECDLP* according to $A_v = \alpha_v \cdot D$. Thus,

$\mathcal{A}$ is impossible to fabricate $\mathcal{VK}_{uv}$.

$$ii) A_v = \mathcal{CR}_v + \mathcal{H}(ID_v || \mathcal{CR}_v || A_v || \mathcal{X}_v || N_v)(\omega_v + \omega_v^* + \alpha_v)$$

To execute such calculation, it is vital for $\mathcal{A}$ to extract the value of α_u , α_v , and $(\omega_v + \omega_v^*)$ to get the necessary elements from the following equations:

$$\begin{aligned} A_u &= \alpha_u \cdot D, \\ A_v &= \alpha_v \cdot D, \\ \Omega_v^\circ &= (\omega_v + \omega_v^*) \cdot D. \end{aligned}$$

To solve such an equation is equivalent to solving the *HECDLP* three times, which proves it is unable for $\mathcal{A}$ to falsify $\mathcal{DRN}_v$'s signature A_v .

On balance, if the adversary $\mathcal{A}$ attempts to impersonate $\mathcal{DRN}_v$, complex mathematical operations are required. This mathematical operation is equivalent to calculating the discrete logarithm problem of hyperelliptic curves at least four times. Therefore, the proposed scheme provides protection against drone impersonation attacks.

5) Replay Attack: We assume that the adversary $\mathcal{A}$ aims to intercept the communication between $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$, attempting to capture and replay the message Ψ_1 in the open channel:

$$\Psi_1 = \{\mathcal{E}_{A_v}(ID_u, N_u, A_u, \mathcal{CR}_u), \mathcal{X}_u, \Omega_u, \Omega_u^\circ, A_u\}.$$

After receiving the replayed message from $\mathcal{A}$, $\mathcal{DRN}_v$ decrypts the message with its own private key to get the nonce N_u and checks its freshness and legitimacy. Since this message is a replayed message, its nonce N_u cannot pass $\mathcal{DRN}_v$'s test, so the attack is unsuccessful.

6) Node Capture Attack: Suppose adversary $\mathcal{A}$ wants to generate the following session key:

$$\mathcal{SK}_{uv} = \mathcal{H}(\mathcal{CR}_u || N_u || \lambda_{uv} || \mu_{uv} || N_v || \mathcal{CR}_v).$$

It is now necessary to consider the capabilities possessed by the adversary. It is possible for $\mathcal{A}$ to capture and obtain a UAV's memory including the long-term private key, as well as other parameters, belonging to one side the communication parties $\mathcal{DRN}_u$ and $\mathcal{DRN}_v$. The following scenario illustrates the subsequent actions taken by $\mathcal{A}$ upon obtaining different key values:

i) $\mathcal{DRN}_u$'s private key α_u is leaked

If $\mathcal{DRN}_u$'s private key α_u is leaked, the aforementioned key will permit access to the encrypted message that was transmitted via the public channel. In this case, $\mathcal{A}$ can acquire NON_{ev} , A_v , $\mathcal{CR}_v$, $\mathcal{VK}_{uv}$ by computing:

$$(NON_{ev}, A_v, \mathcal{CR}_v, \mathcal{VK}_{uv}), ID_v, \Omega_v^\circ, A_v, \mathcal{X}_v = \mathcal{D}_{\alpha_u}(\Psi_2).$$

To generate the session key $\mathcal{SK}_{uv}$, $\mathcal{A}$ must also solve the values of $\mathcal{CR}_u$, N_u , λ_{uv} , μ_{uv} , and N_v . $\mathcal{CR}_u$ can be calculated by equation

$$\mathcal{CR}_u = d_{\mathcal{CA}} + (\mathcal{f}_u + \alpha_u) \mathcal{H}(ID_u || \mathcal{X}_u).$$

Where $d_{\mathcal{CA}}$ is the $\mathcal{CA}$'s private key, and the only way to get it is by using equation $w_{\mathcal{CA}} = d_{\mathcal{CA}} \cdot D$. To perform such a computation is equivalent to solving the *HECDLP*, which is far too complex to be solved by the adversary $\mathcal{A}$.

Then, in the open channel, it is not possible for $\mathcal{A}$ to ascertain the value of N_u , as the message Ψ_1 ought to be decrypted by α_v . Also, to get the value of N_v , $\mathcal{A}$ needs to perform the calculation $(N_v, ID_v) = \mathcal{D}_{\omega_u^*}(NON_{ev})$, the value

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ω_u^* needs to be extracted from the relation $\omega_u^* \cdot D = \mathcal{Q}_u^* = \mathcal{Q}_u^\circ - \mathcal{Q}_u$, and performing such a computation is tantamount to solving the *HECDLP*. With respect to λ_{uv} and μ_{uv} , although $\mathcal{A}$ can obtain the value of μ_{uv} through equation $\mu_{uv} = \alpha_u \cdot A_v = \alpha_u \alpha_v \cdot D$, it is unable to compute the value of λ_{uv} because it lacks the requisite value of ω_v .

ii) $\mathcal{DRN}_v$'s private key α_v is leaked

If $\mathcal{DRN}_v$'s private key α_v is leaked, $ID_u, N_u, A_u, \mathcal{CR}_u$ can be acquired by decrypting:

$$(ID_u, N_u, A_u, \mathcal{CR}_u, \mathcal{X}_u, \mathcal{Q}_u, \mathcal{Q}_u^\circ, A_u = \mathcal{D}_{\alpha_v}(\Psi_1)).$$

In order to generate the session key $\mathcal{SK}_{uv}$, $\mathcal{A}$ must solve the values of $\mathcal{CR}_v, \lambda_{uv}, \mu_{uv}$, and N_v . Except for μ_{uv} , the remaining parameters are not available, so $\mathcal{A}$ cannot compute $\mathcal{SK}_{uv}$ under this circumstance.

Hence, in our proposed scheme, even if it is assumed that the adversary has knowledge of one of the long-term private keys of the two UAVs, this adversary cannot retrieve the session key $\mathcal{SK}_{uv}$ unless they can effectively break the hardness of the *HECDL* problem.

7) Ephemeral Secret Leakage (ESL) Attack: If either $\mathcal{DRN}_u$'s or $\mathcal{DRN}_v$'s temporary private key ω_u or ω_v is leaked, the adversary $\mathcal{A}$ will not be able to unravel any of the parameters that is necessary to generate $\mathcal{SK}_{uv}$ except for λ_{uv} . Therefore, it can be concluded that the proposed scheme is also resilient against ephemeral secret leakage attacks.

8) Malicious Drone Deployment Attack: Consider a scenario where the adversary $\mathcal{A}$ attempts to infiltrate an established network by deploying a fraudulent drone, denoted as $\mathcal{DRN}_{\text{fraud}}$. The adversary's modus operandi involves:

- Assigning a fake identity, ID_{fraud} , and a randomly generated private key, α_{fraud} .
- Deriving the public identity as ID_{fraud} , and computing: $A_{\text{fraud}} = \alpha_{\text{fraud}} \cdot D$.
- Choosing $\mathcal{K}_{\text{fraud}}$ and computing $\mathcal{X}_{\text{fraud}}: \mathcal{X}_{\text{fraud}} = (\mathcal{K}_{\text{fraud}} + \alpha_{\text{fraud}}) \cdot D$.
- Generating a fake certificate for $ID_{\text{fraud}}: \mathcal{CR}_{\text{fraud}} = d_{\text{fraud}} + (\mathcal{K}_{\text{fraud}} + \alpha_{\text{fraud}}) \mathcal{H}(ID_{\text{fraud}} || \mathcal{X}_{\text{fraud}})$.
- Subsequently, loading the dataset $(ID_{\text{fraud}}, \mathcal{CR}_{\text{fraud}}, A_{\text{fraud}}, \alpha_{\text{fraud}}, \mathcal{X}_{\text{fraud}})$ into $\mathcal{DRN}_{\text{fraud}}$'s memory.

In practicality, to authenticate $\mathcal{DRN}_{\text{fraud}}$ with a legitimate certificate, the intruder needs $d_{\mathcal{CA}}$, derivable from $w_{\mathcal{CA}} = d_{\mathcal{CA}} \cdot D$. However, manipulating this relationship is as challenging as solving an *HECDLP*. Consequently, among other safeguards, our scheme defends the system against Malicious Drone Deployment Attacks.

9) Anonymity Preservation: In our proposed scheme, the principle of anonymity is upheld. The ciphertext Ψ_1 does not directly contain the information. The drone identity can only be derived from the ciphertext as follows:

$$(ID_u, N_u, A_u, \mathcal{CR}_u, \mathcal{X}_u, \mathcal{Q}_u, \mathcal{Q}_u^\circ, A_u = \mathcal{D}_{\alpha_v}(\Psi_1)).$$

In order to obtain the drone identity, the private key, α_v , must be used. This requires the solution of the hyperelliptic curve discrete logarithmic problem once. Furthermore, it is challenging to retrieve ID_u from A_u or $\mathcal{CR}_u$ due to the

unidirectional nature of the hash function. Consequently, it can be concluded from the aforementioned argument that the proposed scheme effectively guarantees anonymity.

10) Cloning Attack: In a drone cloning attack, an adversary attempts to replicate or mimic the identity of a legitimate drone, $\mathcal{A}$ obtaining the identity information of the legitimate drone, which includes the drone's identifier ID_i , certificate $\mathcal{CR}_i$, and other authentication credentials.

Nevertheless, as a consequence of the fact that in our scheme a provisional set of session keys ω_i and ω_i^* is selected for each session, such an update renders our scheme resistant to cloning attacks.

11) De-synchronization Attack: During the communication process between the sender $\mathcal{DRN}_u$ and the receiver $\mathcal{DRN}_v$, it is possible for $\mathcal{A}$ to prevent the simultaneous updating of secret information. In order to prevent this attack, the receiving drone $\mathcal{DRN}_v$ is equipped with the capability to store the previous message in the database, accompanied by its nonce. In the event that $\mathcal{A}$ transmits the message Ψ_1 to $\mathcal{DRN}_v$, the latter proceeds to decrypt the parameters, and assesses the freshness of N_u . This process enables the receiving drone to evade the de-synchronization attack.

12) DoS Attack:

The solution is not susceptible to denial-of-service (DoS) attacks. To illustrate, if $\mathcal{DRN}_v$ transmits Ψ_2 , where $\Psi_2 = \{\mathcal{E}_{A_u}(NON_{ev}, A_v, \mathcal{CR}_v, \mathcal{VK}_{uv}), ID_v, \mathcal{Q}_v^\circ, A_v, \mathcal{X}_v\}$ to $\mathcal{DRN}_u$, the initial step is to decrypt Ψ_2 and then NON_{ev} to get N_v , subsequently verifying the freshness of the nonce N_v . If the nonce is no longer fresh, the DoS attack is effectively resisted.

VI. PERFORMANCE ANALYSES OF RELATED SCHEMES

In this part, from the view of the computation consumed costs and communication consumed costs, we give the performance comparative analysis between our scheme and six contemporary state-of-the-art schemes.

A. Computational Costs Comparison

To give a fair computation costs comparison, as a widespread way, we choose the available tested results from existing works. As shown in Table III [18], [40]-[44], the processing time of several dominant operation is: hyperelliptic curve scalar multiplication (HCSM), 0.48ms [38], [42], elliptic curve scalar multiplication (ECSM), 1.206ms [39], [41], executing a secure PUF operation, 1.12ms [43], [47], and executing a fuzzy extractor generation/reconstruction takes 3.46ms. Further, combined with Table IV, it demonstrates our proposed scheme's superiority in aggregated computational costs across six AKA schemes, and confirms the proposed scheme's viability for latency-sensitive applications.

One may notice that although the performance of scheme [18] can be better than the proposed scheme, it is vulnerable to certain threats including ESL attacks and Type-I node capture attacks, the proposed scheme is much more efficient in terms of computational cost with robust security. As shown in Fig. 4, our scheme has the second lowest computational cost, with about 0.5 ms more than the scheme of Muhammad et al. [18], and is

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significantly better than the remaining five schemes.

TABLE III
THE RUNNING TIME FOR EACH OPERATION

Item	Descriptions	Time(ms)
T_{hm}	Hyperelliptic curve scalar multiplication	4.8×10^{-1}
T_{em}	Elliptic curve scalar multiplication	1.206
T_{ea}	Elliptic curve addition	2.88×10^{-2}
T_{AE}	The time of AES's encryption	3.53×10^{-2}
T_h	Execute a secure hash function	1.2×10^{-2}
T_E	The time of HECC's encryption	1.1×10^{-3}
T_D	The time of HECC's decryption	3.5×10^{-4}
T_P	Secure PUF operation	1.12
T_{fe}	Fuzzy extractor generation/reconstruction	3.46

TABLE IV
COMPLEXITY OF COMPUTATIONAL COSTS

Schemes	Formal Computational Cost
M. A. Khan et al. [18]	$14T_{hm} + 11T_h + 3T_E + 3T_D$
Gope et al. [45]	$10T_h + 2T_P + 2T_{fe}$
Yahuza et al. [46]	$6T_{em} + 6T_h + 10T_{ea}$
Bera et al. [47]	$12T_{em} + 12T_h + 4T_{ea}$
K. Huang et al. (1) [48]	$8T_{em} + 9T_h + 2T_{ea} + 2T_P$
K. Huang et al. (2) [48]	$16T_{em} + 14T_h + 4T_{ea} + 2T_P + 2T_{AE}$
Our scheme	$15T_{hm} + 10T_h + 3T_E + 3T_D$

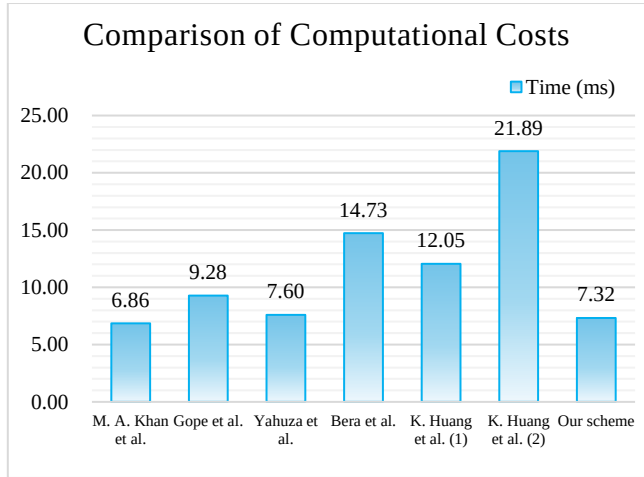


Fig. 4 Comparison of computational costs.

B. Communication Costs Comparison

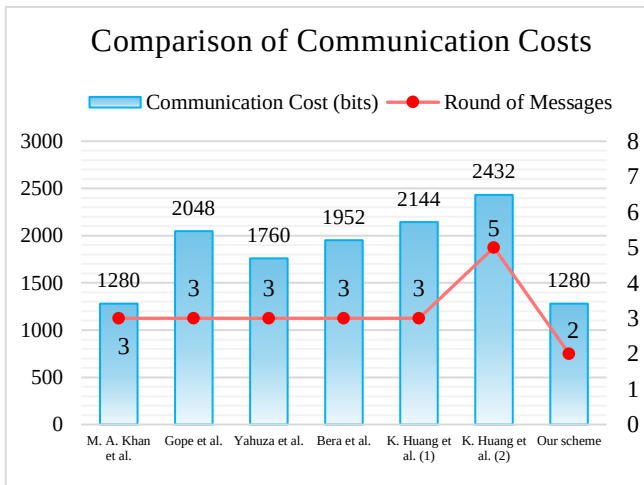


Fig. 5 Comparison of communication costs.

To compare communication costs, we quantify the overhead

based on the number of messages exchanged and their sizes. For each scheme, we calculate the total length and round of messages involved in the authentication and key agreement process. From Fig. 5, we can see that our scheme requires the fewest round of messages and communication cost in bits, saving nearly 50% of communication cost compared with K. Huang et al.'s scheme (2) [48], thereby reducing computational cost, memory requirements, and energy consumption. Moreover, our scheme only takes two rounds of communication to achieve AKA after optimization. The results demonstrate our scheme's efficiency in minimizing communication. It is obvious to see that our proposed scheme achieves the comparable results.

VII. CONCLUSION

AKA mechanisms are crucial for ensuring the security of UAV communications. This paper has presented a new HECC-based AKA scheme for resource-constrained UAV systems, which ensures robust UAV mutual authentication with precise identity verification, while facilitating secure session key establishment to maintain communication confidentiality. By leveraging HECC, the proposed scheme has provided robust security against threats including ESL attacks and node capture attacks under the eCK adversarial model. The scheme's security analysis has demonstrated its resilience against eCK adversarial threats. Performance comparisons with six schemes have further demonstrated its efficiency, with significant reductions in computational costs and communication costs. It has been shown that less communication rounds can still achieve a balance between security assurance and operational efficiency in resource-limited aerial environments. In our future work, we will explore the integration of quantum-resistant algorithms to counter post-quantum threats and adaptive AKA mechanisms for dynamic and multi-domain UAV networks, broadening its applicability across diverse operational scenarios.

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